



Comparing protein sequence alignment score networks

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September 2026

Abstract

Sequence-similarity networks encode protein sequences as vertices and selected pairwise similarities as edges. Resulting topology depends on protein sample, score method, and edge rule. We implemented a reproducible pipeline and tested these choices on 960 Pfam domains from 24 families and 8 clans. A factorial design generated 900 balanced collections and 21,600 weighted graphs using exact local alignment or BLAST, three edge rule families, and Louvain community detection. Adjusted mutual information (AMI) measured recovery of withheld Pfam labels. At matched 2% edge density, median absolute family NMI difference between methods was 0.0034. Equal score percentiles produced unequal densities. Score permutation reduced family AMI at reference setting from 0.936 to 0.009. Results quantify sampling and edge selection effects in protein network analysis.

1 Introduction

We may consider protein as a finite ordered sequence of amino-acid residues and intuitively would expect proteins with similar sequences to be similar in some biological sense. Protein alignment is then the family of techniques used to estimate "similarity" from amino acid sequences alone. Typically protein sequences are very long and instead we focus on protein domains - shorter, contiguous subsequences of a protein.

For any given pair of protein domain sequences of comparable length, pairwise alignment inserts gaps into two such sequences so that residues judged to correspond occupy shared positions; a gap marks residues present in one sequence and absent in the other, an indel whose direction pairwise alignment does not determine. A substitution matrix scores each aligned residue pair and gap costs penalise indels, so total score rewards correspondence and penalises divergence. Global alignment scores two sequences over their full length; local alignment selects one region of each sequence maximising total score, and so tolerates unrelated flanking segments. Comparison at domain level uses local alignment. Several algorithms produce such a score, differing in computational cost and in which pairs they report and is further discussed in Section 2

Pair scores are then used to define a sequence-similarity network (SSN) where vertices represent sequences, and selected pairs form weighted edges. SSNs support exploration and clustering of protein families [Atkinson et al., 2009, Gerlt et al., 2015, Enright et al., 2002]. Sequences alone do not determine an SSN. Three choices intervene: the collection of proteins considered, the alignment algorithm to score pairs, and the graph rule that promotes scores to edges. Threshold choice has no accepted universal value, and reported topology shifts with it [Apeltsin et al., 2011, Hornung and Terrapon, 2023].

We then seek to ask how graphs built from different protein collections, alignment algorithms, and graph construction rules compare and how well, if at all, do these graphs recover known biological structure.

This study permutes those effects on a recorded panel of 960 Pfam seed domains from 24 families and 8 clans. A factorial design samples 900 balanced collections; two scoring methods and 12 edge settings yield 21,600 weighted graphs. Community detection runs without labels, and agreement with withheld Pfam family and clan labels quantifies recovery.

2 Alignment algorithms and scoring

2.1 Exact local alignment

Let $x = x_1 \dots x_p$ and $y = y_1 \dots y_q$ be amino-acid sequences. Substitution matrix $a(\cdot, \cdot)$ assigns score $a(x_i, y_j)$ to an aligned residue pair. BLOSUM62 is a 20×20 substitution matrix estimated from conserved blocks of aligned proteins, its entries scaled log-odds of observed against chance residue pairing [Henikoff and Henikoff, 1992].

We use affine gap cost [Gotoh, 1982], where opening a gap costs $g_o > 0$ and each further gap residue costs g_e , with $0 < g_e < g_o$, so a gap of length ℓ costs $g_o + (\ell - 1)g_e$. One long gap therefore costs less than several short gaps of equal total length, since a single multi-residue indel occurs more readily than several isolated ones. We take $g_o = 11$ and $g_e = 1$, conventional values for BLOSUM62.

The Smith–Waterman algorithm finds an exact optimal local alignment by dynamic programming [Smith and Waterman, 1981]. Gotoh’s formulation attains affine gap costs in $O(pq)$ time by carrying separate states for gapped and ungapped extensions,

$$\begin{aligned} E_{ij} &= \max\{H_{i-1,j} - g_o, E_{i-1,j} - g_e\}, \\ F_{ij} &= \max\{H_{i,j-1} - g_o, F_{i,j-1} - g_e\}, \\ H_{ij} &= \max\{0, H_{i-1,j-1} + a(x_i, y_j), E_{ij}, F_{ij}\}. \end{aligned}$$

Here H_{ij} is best score of a local alignment ending with x_i aligned to y_j ; E_{ij} is best score ending with x_i aligned to a gap in y ; F_{ij} is best score ending with a gap in x aligned to y_j . Boundary values are $H_{i0} = H_{0j} = 0$, with gap states unavailable at a boundary set to $-\infty$. Zero inside H_{ij} lets an alignment restart, which yields a local rather than global optimum. Under this recurrence a gap of length ℓ costs $g_o + (\ell - 1)g_e$, and local score is

$$s(x, y) = \max_{i,j} H_{ij}.$$

Biopython’s PairwiseAligner [Cock et al., 2009] computed these scores in local mode with BLOSUM62, $g_o = 11$, and $g_e = 1$.

2.2 Heuristic search

The BLASTP algorithm avoids exhaustive dynamic programming. It indexes short words, locates high-scoring word matches between query and subject, and extends only those seeds into local alignments [Altschul et al., 1990, Camacho et al., 2009]. Cost per pair falls well below $O(pq)$. A weakly similar pair may yield no seed and therefore no reported alignment, so absence from BLAST output records failure to reach a reporting threshold rather than a score of zero.

Raw score S carries units of its scoring system. Bit score rescales it to a scale independent of scoring system,

$$S' = \frac{\lambda S - \log K}{\log 2},$$

where λ and K are Karlin–Altschul parameters fixed by substitution matrix, gap costs, and background residue frequencies [Karlin and Altschul, 1990]. Graph weights used S' .

BLAST runs used BLOSUM62 and nominal gap costs 11 and 1, matching exact local alignment of Section 2.1, so both methods score aligned residue pairs from one substitution matrix. BLAST additionally applies composition-based statistics in mode 2, which rescales substitution scores separately for each query–subject pair according to that pair’s residue composition. Rescaling suppresses high scores arising from compositional bias rather than from shared ancestry, for instance between two sequences both rich in a single residue.

BLAST reports an alignment only when its expect value falls below a threshold. For an alignment of raw score S and bit score S' ,

$$E = Km'n' e^{-\lambda S} = m'n' 2^{-S'},$$

where m' and n' are effective query and database lengths after correction for alignments truncated at sequence ends, and λ , K are as above. Both forms agree because $2^{-S'} = Ke^{-\lambda S}$, the bit score being defined precisely so that expect value takes this form. Hence E counts alignments expected to reach score S by chance alone in a search of that size, and small E marks a score unlikely to arise at random. Since E scales with $m'n'$, one value of E denotes different degrees of similarity in databases of different size.

Runs used $E = 10$ under standard BLAST default. That value is permissive by design, far above thresholds near 10^{-3} conventionally used to claim homology, and it exists to surface marginal hits rather than to establish significance. Permissiveness is apt here since edge selection happens downstream under rules of Section 3.4, leaving E to determine only which pairs carry a score at all.

BLAST searches are directional and may return several high-scoring segment pairs (HSPs) for one query–subject comparison. HSP of highest bit score represented each ordered pair, and highest score across reported directions represented each unordered pair. BLAST reported 19,376 of 460,320 master pairs, or 4.21%.

2.3 Comparability of the two scoring systems

Both methods used BLOSUM62 and nominal gap parameters 11 and 1. Two differences remain. BLAST charges an extension cost for every gap residue, so a gap of length ℓ costs $g_o + \ell g_e$, exceeding cost under exact alignment by g_e at every gap. Composition-based statistics further adjust BLAST substitution scores per sequence pair, while exact local alignment applies one fixed matrix throughout.

Each method therefore ran at its own standard default. Method comparisons in this report control vertex set and edge count (Section 3.4) rather than assert identical scoring systems.

2.4 Learned alignment

DEDAL is a pretrained neural model that learns sequence representations and alignment parameters from protein data, returning a Smith–Waterman-style score [Llinares-López et al., 2023]. Pretrained model was verified on pilot data. Measured runtime exceeded available compute budget for all-versus-all scoring of master panel, so scaled results use exact local alignment and BLAST only.

3 Methods

3.1 Networks from pair scores

For collection C , let V be its vertex set, $n = |V|$, $\mathcal{P}(V) = \{\{i, j\} : i, j \in V, i \neq j\}$, and $P = |\mathcal{P}(V)| = \binom{n}{2}$. Method m reports score s_e^m for each pair in available set $A_m \subseteq \mathcal{P}(V)$; pairs outside A_m stay missing and never receive an imputed value. Edge setting θ selects $E_{m,\theta} \subseteq A_m$, giving weighted graph

$$G(C, m, \theta) = (V, E_{m,\theta}, w_m), \quad w_m(e) = s_e^m,$$

with realised density $\delta = |E_{m,\theta}|/P$, fraction of possible pairs that became edges. Term *realised* marks δ as an outcome rather than an input, since top- k and percentile rules fix no edge count in advance and target density can fall short of its request. Every input sequence remains a vertex; sequences with no selected pair become isolates.

Each vertex carries a stable identifier formed from its UniProt accession with domain start and end coordinates, for example A0A0H3GJZ1_27_153. Lexicographic order of identifiers resolves ties wherever a selection would otherwise be ambiguous, making every procedure below deterministic.

Path from collection to reported quantity is

$$C \longrightarrow \{s_e^m : e \in A_m\} \longrightarrow G(C, m, \theta) \longrightarrow U(C, m, \theta) \longrightarrow \text{AMI}(U, V^*),$$

where U is a community partition detected without labels and V^* is withheld Pfam partition. Remaining subsections define each arrow. Master panel and sampling design fix C , edge rules fix θ , and community detection with agreement measures fixes U and final comparison.

3.2 Master panel

The Pfam database defines families of homologous domains using curated seed alignments and profile hidden Markov models; clans group families with evidence of common ancestry [Mistry et al., 2021].

Panel selection follows a rejection sampling procedure over Pfam clans, run under seed 20260828. The procedure first reads official family–clan catalogue, discards unassigned families and clans holding fewer than three catalogue families, then shuffles sorted clan list. Within each clan in that order it shuffles sorted family list and inspects families in turn, accepting those of InterPro type *domain* or *family* carrying at least 40 reported seed domains. A clan is retained once three families are accepted, and skipped when fewer prove eligible. Selection stops at 8 retained clans. One random stream drives every shuffle, so seed alone reproduces the panel. Trace table records every family inspected together with its acceptance or rejection. Pseudocode of the full procedure may be found in Appendix A.

Each selected seed alignment is filtered to unique sequences over 20 standard amino acids with consistent coordinates. Deterministic farthest-point sampling then selects 40 representatives per family. First sequence has length nearest family median. Given selected set S and remaining set R , next sequence is

$$x^* = \arg \max_{x \in R} \min_{y \in S} d(x, y), \quad d(x, y) = 1 - \text{identity}(x, y),$$

where identity counts matching residues over jointly occupied seed alignment positions. Identifier order resolves ties. This maximises within-family sequence diversity, so recovery results are conservative relative to a sample of each family weighted by frequency. Alignment gaps are removed before scoring. Panel contains 960 domains, 24 families, 8 clans, and $\binom{960}{2} = 460,320$ unordered pairs.

3.3 Repeated collections

Each collection is a balanced hierarchical draw. For condition (c, f, r) , sample c clans uniformly without replacement, then f families within each selected clan, then r domains within each selected family. Collection size is $n = cfr$, with equal family sizes by construction. Factor levels are

$$c \in \{2, 4, 8\}, \quad f \in \{2, 3\}, \quad r \in \{10, 20, 40\}.$$

Cartesian product gives $3 \cdot 2 \cdot 3 = 18$ conditions, and 50 seeded replicates per condition give $18 \cdot 50 = 900$ collection declarations. A stable hash of base seed, condition, and replicate index generates each replicate seed, so extending the design leaves earlier collections unchanged. Pfam labels determine sampling strata and are then withheld; they enter no score, edge rule, or community detection.

Conditions requesting most of the panel can repeat a complete node set across replicates despite distinct seeds. A uniqueness audit hashes sorted node identifiers within each condition. 4 conditions contain duplicates; their 200 declarations remain in audit tables and figures but leave distributional summaries. Primary distributions therefore use 700 collections from 14 conditions with 50 distinct node sets each. All 4 excluded conditions request $r = 40$, so exclusion falls entirely on largest domain count and composition effects reported below are conservative. Two score methods and 12 edge settings generate 21,600 graphs before this filter.

3.4 Edge rules

Recall vertex set V of collection C , possible pairs $\mathcal{P}(V)$ with $P = |\mathcal{P}(V)|$, and available set $A_m \subseteq \mathcal{P}(V)$ of pairs that method m scores. Edge setting θ is a rule family together with its parameter, taking one of forms (top, k) , (pct, p) , or (dens, d) defined below.

For $i \in V$, available neighbours are $A_m(i) = \{j \in V : \{i, j\} \in A_m\}$. Let $N_k^m(i)$ be the first $\min\{k, |A_m(i)|\}$ elements of $A_m(i)$ under descending score with ties broken by ascending identifier. For $e \in A_m$, let $r_e^m \in (0, 1]$ be ascending rank of s_e^m with ties averaged and divided by $|A_m|$ and q_e^m be descending rank with $q_e^m = 1$ at highest score. Let $b_m(d) = \min\{|A_m|, \lceil dP \rceil\}$. Three rule families are

$$\begin{aligned} E_{m,k}^{\text{top}} &= \{e = \{i, j\} \in A_m : j \in N_k^m(i) \text{ or } i \in N_k^m(j)\}, \\ E_{m,p}^{\text{pct}} &= \{e \in A_m : r_e^m \geq p\}, \\ E_{m,d}^{\text{dens}} &= \{e \in A_m : q_e^m \leq b_m(d)\}. \end{aligned}$$

$E_{m,k}^{\text{top}}$ (top, k) lets each vertex nominate up to k strongest available neighbours and retains undirected union, so retained edge count grows with n rather than with $\binom{n}{2}$ [Camoglu et al., 2006]. $E_{m,p}^{\text{pct}}$ (pct, p) retains upper tail of scores a method reported, so it acts on A_m ; it coincides with target density $1 - p$ exactly when $A_m = \mathcal{P}(V)$, and otherwise yields a sparser graph in proportion to missing coverage. $E_{m,d}^{\text{dens}}$ (dens, d) requests a common global edge budget $\lceil dP \rceil$ on identical vertex sets, with shortfall $\max\{0, \lceil dP \rceil - |A_m|\}$ recording a budget a method could not supply. Parameters are

$$k \in \{1, 2, 5, 10\}, \quad p \in \{0.90, 0.95, 0.98, 0.99\}, \quad d \in \{0.005, 0.01, 0.02, 0.05\},$$

giving 12 settings per method.

3.5 Communities and agreement

Communities were detected using weighted Louvain optimisation [Blondel et al., 2008], as implemented in NetworkX [Hagberg et al., 2008]. This heuristic repeatedly moves vertices and aggregates communities to increase weighted modularity

$$Q = \frac{1}{2W} \sum_{i,j} \left(w_{ij} - \gamma \frac{k_i k_j}{2W} \right) \mathbf{1}\{u_i = u_j\}.$$

For vertices i and j , w_{ij} is their edge weight when $\{i, j\} \in E$, and zero otherwise. Weighted degree is $k_i = \sum_j w_{ij}$, and total edge weight is $W = \frac{1}{2} \sum_{i,j} w_{ij}$. Parameter $\gamma > 0$ controls community resolution; all experiments use $\gamma = 1$. Variable u_i denotes the community assigned to vertex i , hence indicator $\mathbf{1}\{u_i = u_j\}$ equals one when i and j share a community.

Choose one of n vertices uniformly. Let U be its detected community and V^* its withheld Pfam label. Let n_{uv} count vertices assigned to community u with label v , $n_u = \sum_v n_{uv}$, and $n_v = \sum_u n_{uv}$. Joint and marginal proportions are

$$p_{uv} = \frac{n_{uv}}{n}, \quad p_u = \frac{n_u}{n} = \sum_v p_{uv}, \quad p_v = \frac{n_v}{n} = \sum_u p_{uv}.$$

Define

$$I(U; V^*) = \sum_{u,v} p_{uv} \log \frac{p_{uv}}{p_u p_v}, \quad H(U) = - \sum_u p_u \log p_u, \quad H(V^*) = - \sum_v p_v \log p_v,$$

with $0 \log 0 = 0$; $\log$ denotes natural logarithm. Normalised mutual information and adjusted mutual information are

$$\begin{aligned} \text{NMI}(U, V^*) &= \frac{I(U; V^*)}{[H(U) + H(V^*)]/2}, \\ \text{AMI}(U, V^*) &= \frac{I(U; V^*) - \mathbb{E}[I(U; V^*)]}{[H(U) + H(V^*)]/2 - \mathbb{E}[I(U; V^*)]}. \end{aligned}$$

Expectation averages mutual information over random contingency tables with observed row and column margins fixed. NMI lies in $[0, 1]$ and rises with partition count even under independence, since a finer partition raises $I(U; V^*)$ by chance alone. AMI subtracts that expectation, takes value 0 in expectation under independence, and so permits comparison across settings whose community counts differ [Vinh et al., 2010]. Both metrics used scikit-learn [Pedregosa et al., 2011]. Family and clan labels entered evaluation only after community detection.

3.6 Reproducibility and figure provenance

Stored graph tables carry one row per graph and reference label level, and all reported analyses read those tables. Script `scripts/build_report_assets.py` generates every figure and numerical macro from saved TSV and JSON artifacts. Canonical PNGs are written under `outputs/figures/pfam_large_panel_report/`; identical copies are placed in `report/figures/`. File `report/figure_provenance.tsv` maps each image to its generating script, exact input tables, both paths, and SHA-256 checksum. Report prose imports generated macros from `report/generated_results.tex`, so no numerical result is transcribed by hand.

4 Results

Analysis centers around which edge rule yields communities closest to withheld Pfam structure. Each Louvain partition was compared separately with family and clan assignments. A tested setting is judged by median AMI across 700 distinct primary collections; NMI shows effect of chance correction. Detailed distribution plots use reference condition $(c, f, r) = (4, 2, 20)$: 50 collections, each containing 160 domains from 8 families.

4.1 Graph-rule recovery of Pfam structure

Figure 1 gives median AMI against median realised density. Family recovery increased from top-1 to top-10 under both score methods. Top-10 gave highest tested family AMI: 0.952 for Biopython and 0.928 for BLAST. Corresponding clan AMI was 0.772 and 0.734. Biopython percentile 0.90 gave clan AMI 0.778, exceeding Biopython top-10 by 0.006; its BLAST counterpart gave 0.166. Top-10 was therefore first or within 0.006 of first for both label levels and both score methods.

Top-5 supplied a lower-density operating point. Median family AMI was 0.938 for Biopython and 0.917 for BLAST at densities 4.18% and 3.87%. Median clan AMI was 0.731 and 0.705. Subsequent examples use top-5 to retain fewer edges; both median family AMI values exceed 0.91.

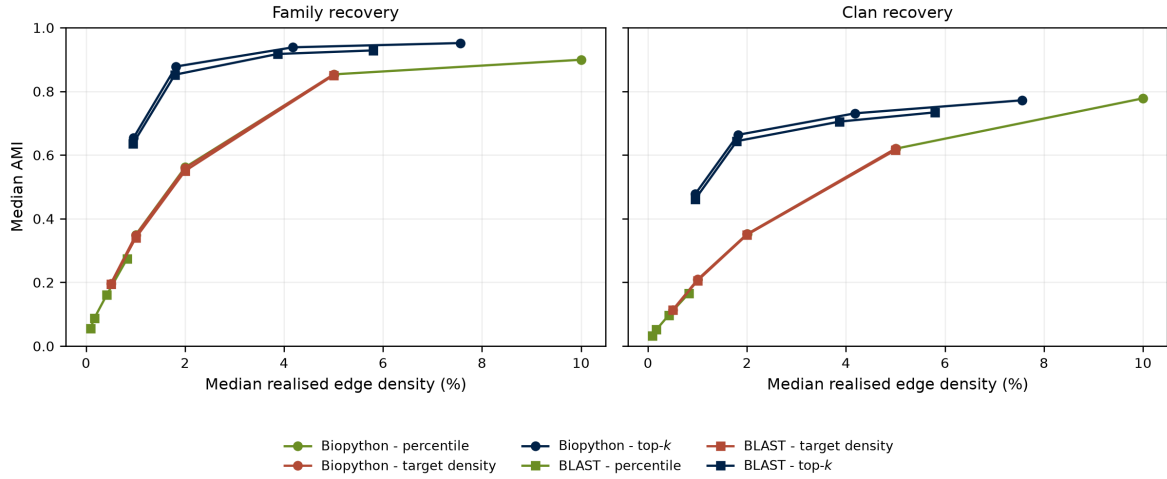


Figure 1: Median AMI against median realised density across 700 primary collections. Panels show family and clan recovery. Lines connect parameter values within each method and rule family.

Figure 2 shows one reference-condition top-5 example. Each panel contains 160 vertices from 8 families. Biopython retained 564 edges in 8 connected components; BLAST retained 558 edges in 7 components. Both family AMI values round to 0.958, although component structures differ.

Each vertex represents one domain; colour denotes withheld Pfam family. Each island is a connected component. Components are packed into a grid, then vertices within each component receive a spring layout. Shared coordinates permit vertex-wise comparison between panels; distance between islands carries no analytical meaning.

4.2 Effect of sparsity on structure recovery

Realised density $\delta = |E|/P$ reads directly as an edge fraction, so $\delta = 0.04$ means 4% of possible pairs are edges. Target density falls short of its request whenever $|A_m| < \lceil dP \rceil$.

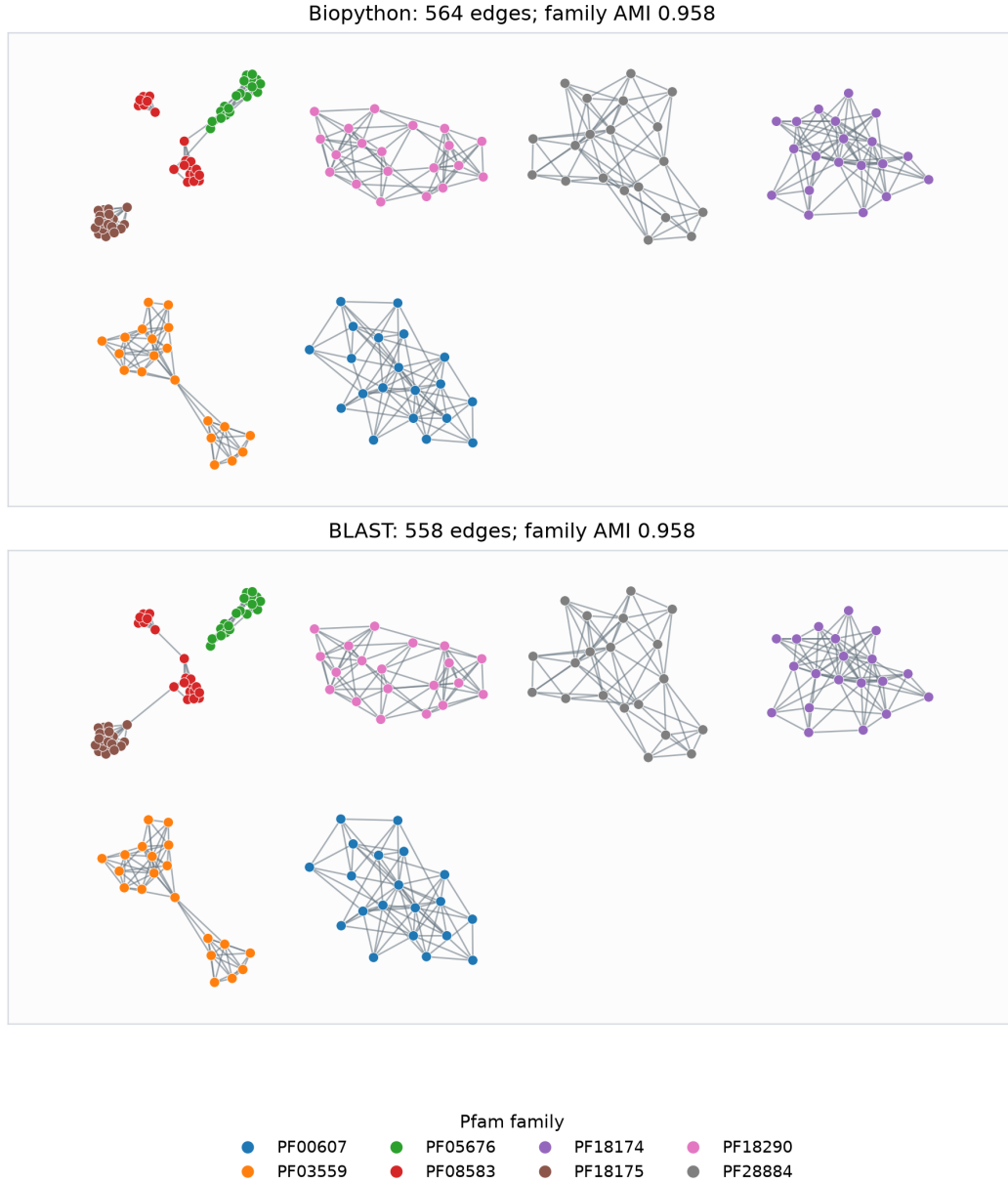


Figure 2: Biopython and BLAST top-5 graphs for collection c4_f2_d20_r001. Colour denotes withheld Pfam family; islands are connected components. Panels share coordinates.

Reference composition fixes vertex count and family mixture. Within this condition, connected-component and Louvain-community counts fell as realised density increased (Fig. 3). At similar density, top- k graphs contained fewer components than percentile and target-density graphs. Top- k allocates local neighbours to every vertex; global rules can concentrate edges among a subset of vertices. Fragmentation split Pfam groups across many inferred communities and reduced AMI.

4.3 AMI as primary agreement measure

NMI and AMI agreed for dense graphs whose inferred community counts approached reference group counts (Fig. 4). Their difference grew under sparse percentile and target-density rules. For BLAST percentile 0.99, median partition contained 150 communities for 8 reference families. Median family NMI was 0.590; median AMI was 0.057. NMI retained agreement produced by many small groups, while AMI subtracted expected agreement conditional on group sizes.

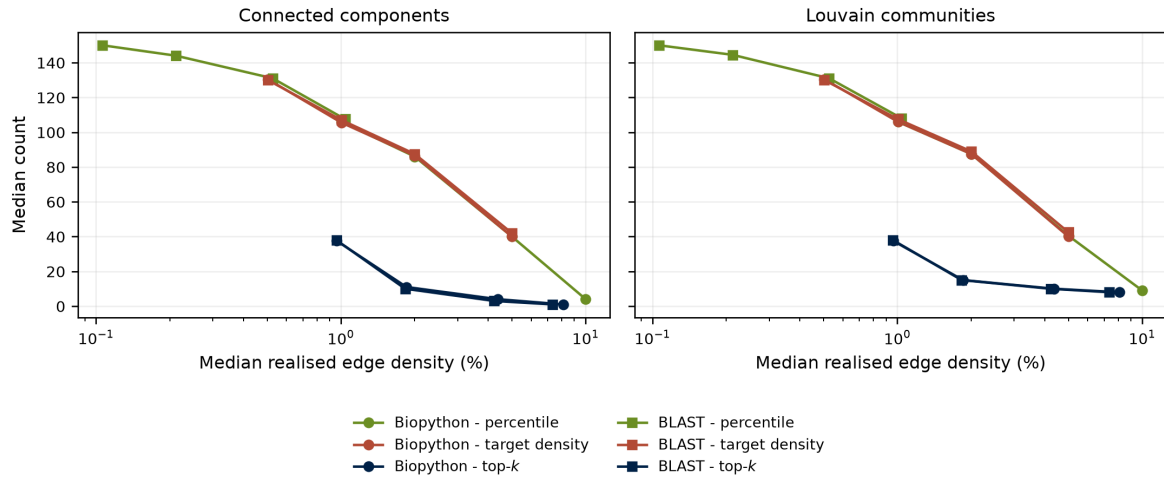


Figure 3: Median component and community counts against realised density for 12 graph settings and 50 reference-condition collections. Colour denotes rule family; marker denotes score method.

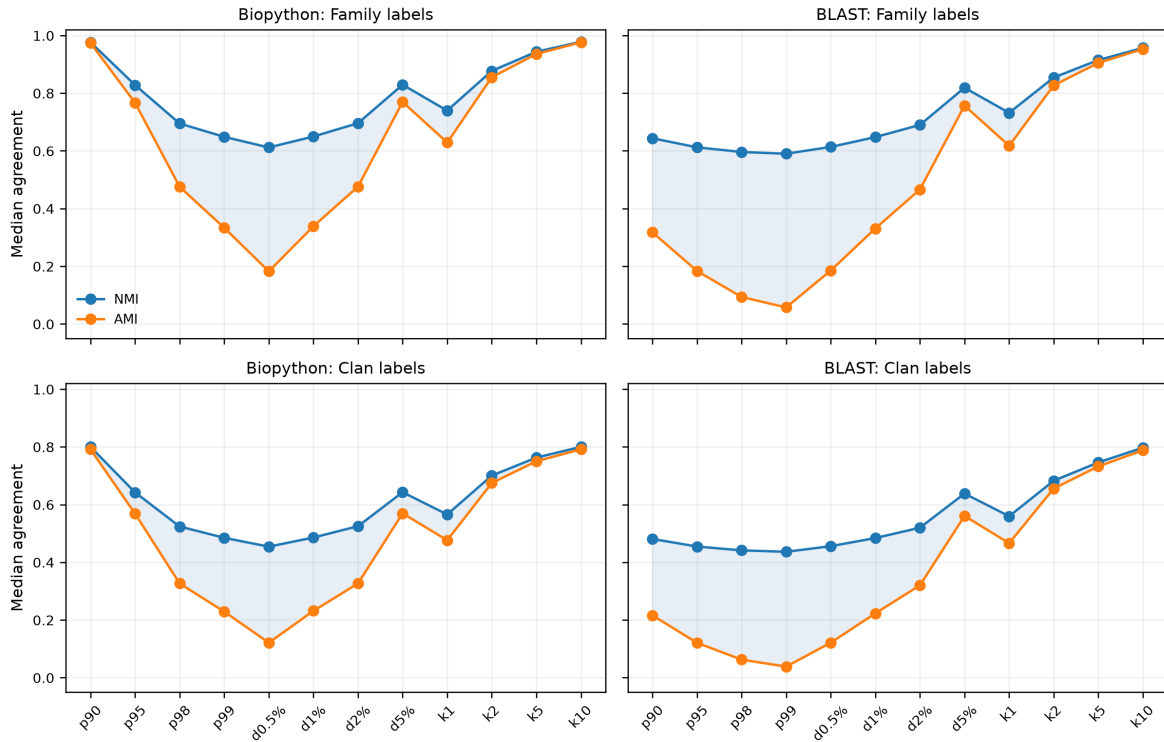


Figure 4: Median NMI and AMI for 12 graph settings in reference condition. Panels separate score method and withheld label level. Shading marks chance correction. Prefixes p , d , and k denote percentile, target density, and top- k .

Figures 5 and 6 show frequency polygons derived from replicate distributions behind Fig. 4. Each of 50 collections contributes one graph to every score-method and graph-setting combination. Agreement values are placed in 25 equal-width bins on $[0, 1]$. Vertical position gives percentage of collection graphs in each bin; frequencies for each curve sum to 100%. AMI moved sparse-rule distributions towards zero. Primary rule comparisons therefore use AMI; NMI remains a bounded description of unadjusted partition agreement.

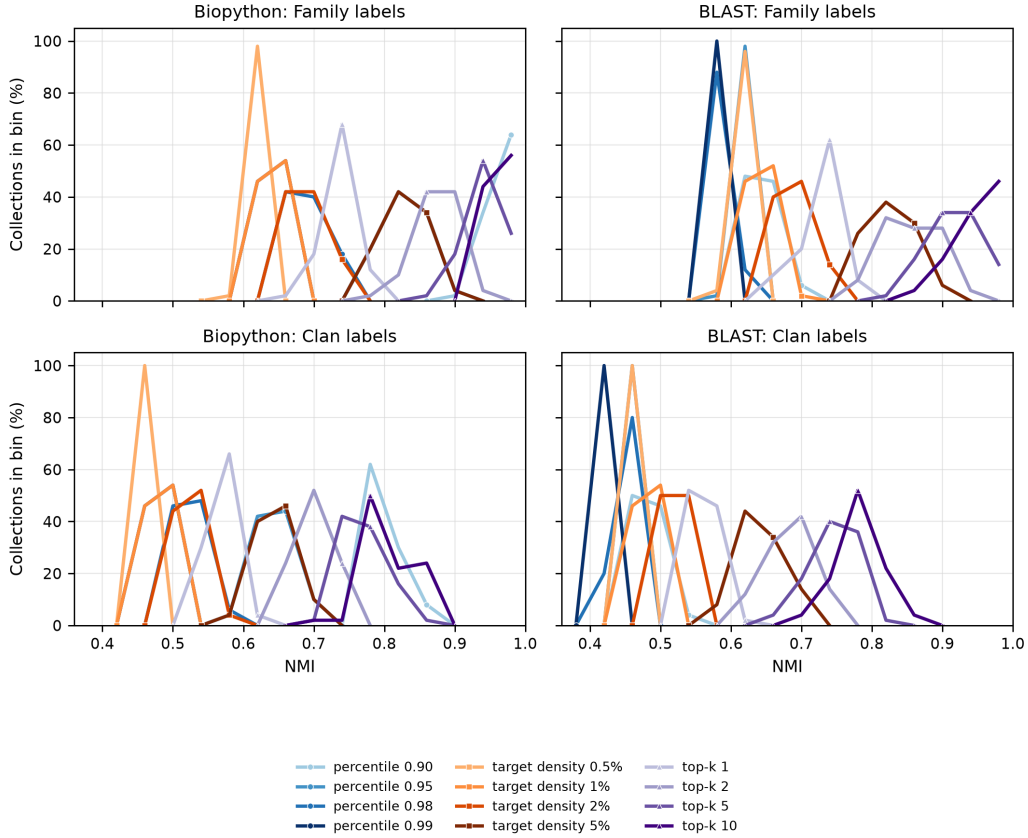


Figure 5: Empirical NMI frequency polygons across 12 graph settings and 50 reference-condition collections. Each curve joins percentages of collection graphs in 25 equal-width NMI bins. Panels separate score method and withheld label level. Line style and marker denote rule family; colour distinguishes parameter values within each family.

4.4 Effect of score method on graph-rule conclusions

Top- k led family recovery under both score methods (Fig. 1). Biopython scored all 460320 master-panel pairs; BLAST reported 19376 pairs, or 4.21%. Among reported pairs, score ranks had Spearman correlation 0.976 (Fig. 7, left). BLAST reports concentrated among pairs with high Biopython scores (Fig. 7, right).

Percentile rules retain a fraction of each method’s available scores. At percentile 0.90, median realised density was 10.00% for Biopython and 0.83% for BLAST (Fig. 8). Equal percentile parameters therefore represented different edge budgets.

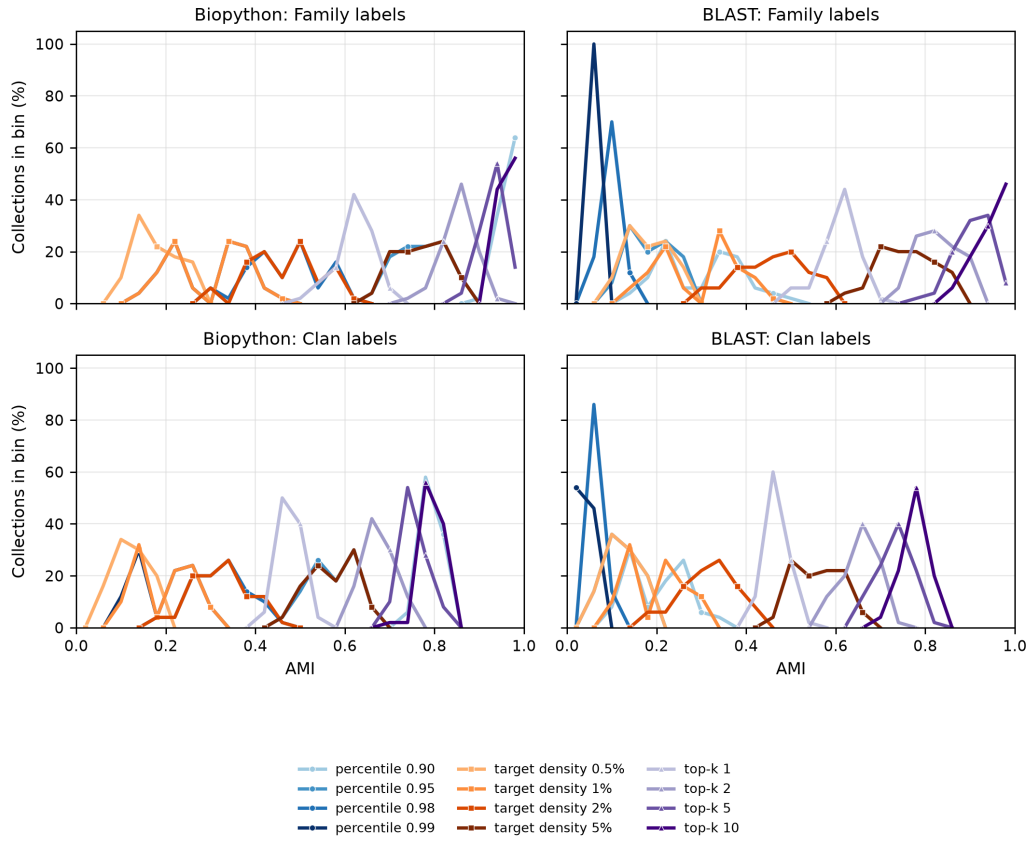


Figure 6: Empirical AMI frequency polygons for graphs in Fig. 5. Panels, curves, bins, and visual encodings follow same design.

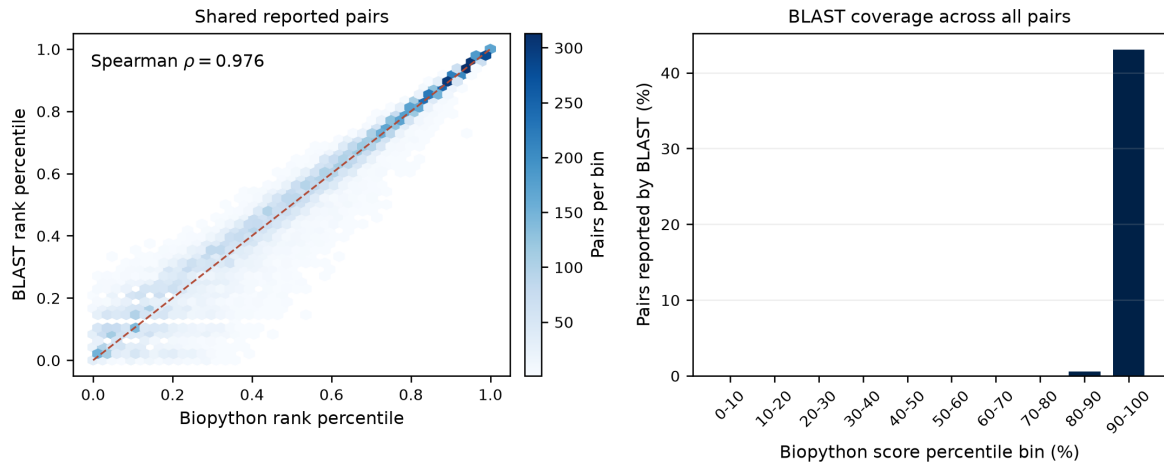


Figure 7: Pair-score comparison on master panel. Left: within-method rank percentiles among 19376 pairs reported by BLAST; colour gives bin count. Right: fraction reported by BLAST within each Biopython score percentile bin across all 460320 pairs.

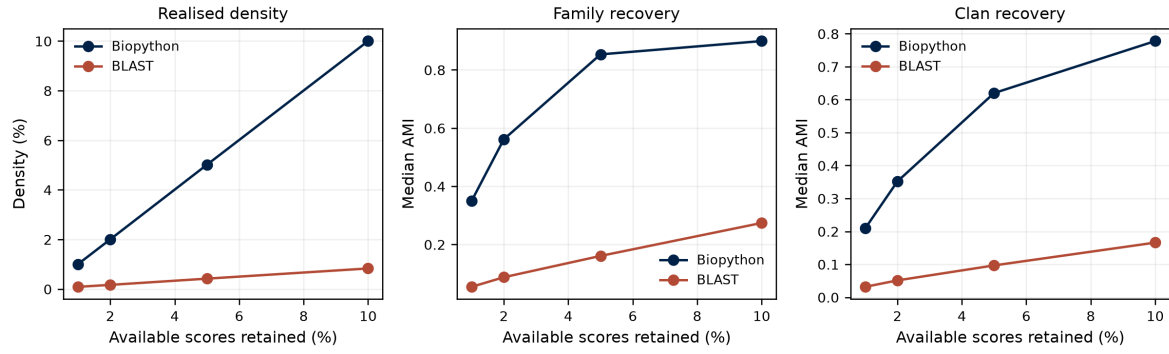


Figure 8: Median density, family AMI, and clan AMI under percentile rules. Horizontal position is fraction of available scores retained within each method. Summaries use 700 primary collections.

Target density rules compare methods on identical vertex sets and requested edge budgets. Budgets 0.5%, 1%, and 2% were fully supplied, while a 5% request exceeded available BLAST pairs in some collections and was therefore excluded from matched budget interpretation. At 2%, paired median absolute family difference was 0.0034 for NMI and 0.0094 for AMI. Signed differences remained close to zero and their spread increased with density (Fig. 9).

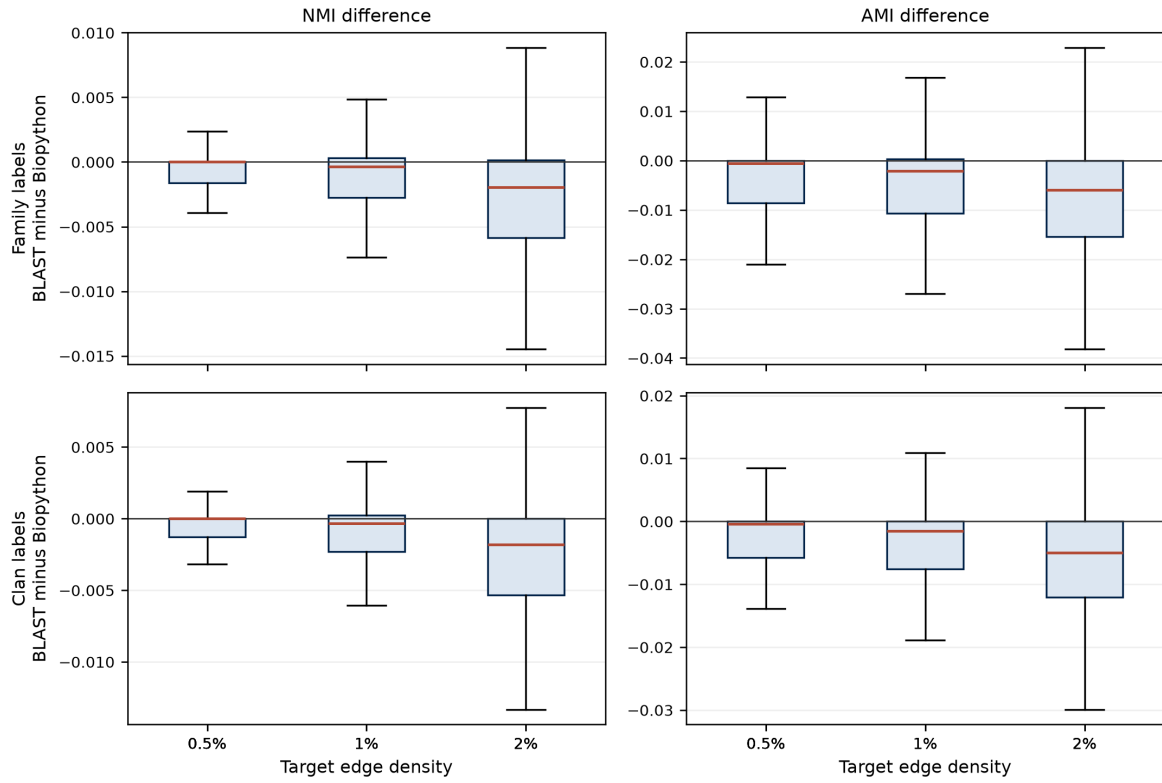


Figure 9: Paired NMI and AMI differences, BLAST minus Biopython, at matched edge densities across 700 collections. Boxes show quartiles and medians; whiskers extend to 1.5 interquartile ranges. Outliers are omitted from display.

4.5 Effect of collection composition under a fixed rule

Top-5 family AMI was evaluated across 14 primary composition conditions, each with 50 collections. Median family AMI ranged from 0.883 to 1.000 for Biopython and from 0.879 to 0.931 for BLAST (Fig. 10). Changes across domain, family, and clan counts were non-monotone. Four saturated design cells remain in audit outputs and do not enter these summaries.

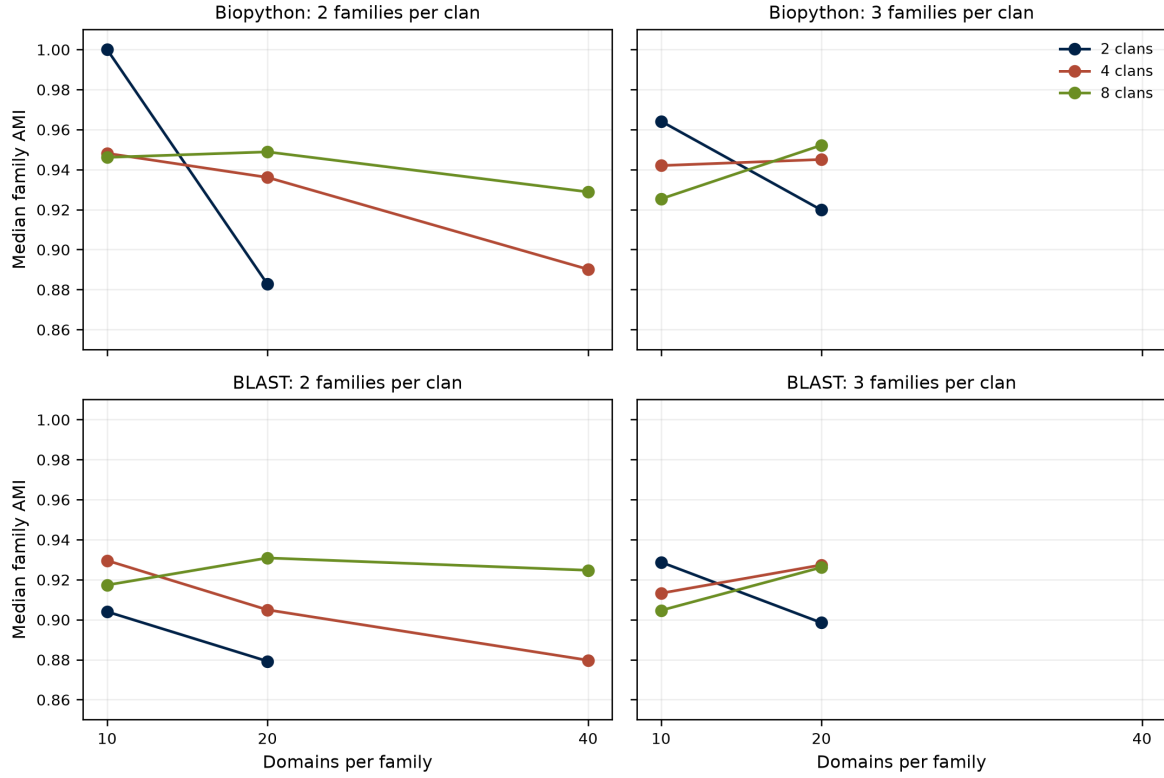


Figure 10: Median family AMI across 14 primary composition conditions under top-5 construction. Colour gives clan count; panels separate score method and families per clan. Lines connect domain counts.

For fixed k , retained edge count grows at most linearly with vertex count n , while possible pair count grows as $\binom{n}{2}$. Realised density therefore decreased with collection size (Fig. 11, left). Median component count also varied with clan and family counts (Fig. 11, right).

4.6 Sensitivity analyses

Score permutation within each collection preserved score distribution and rule parameters while randomising pair identities. In reference condition, Biopython top-5 median family AMI changed from 0.936 to 0.009 (Fig. 12, left). Positive null NMI confirmed chance bias in finite partitions addressed by AMI.

Unweighted Louvain replaced every retained edge weight by one. Reference Biopython top-5 median family AMI was 0.953, compared with 0.936 under native weights (Fig. 12, right). Selected adjacency therefore retained family agreement after removal of edge weight variation for this setting.

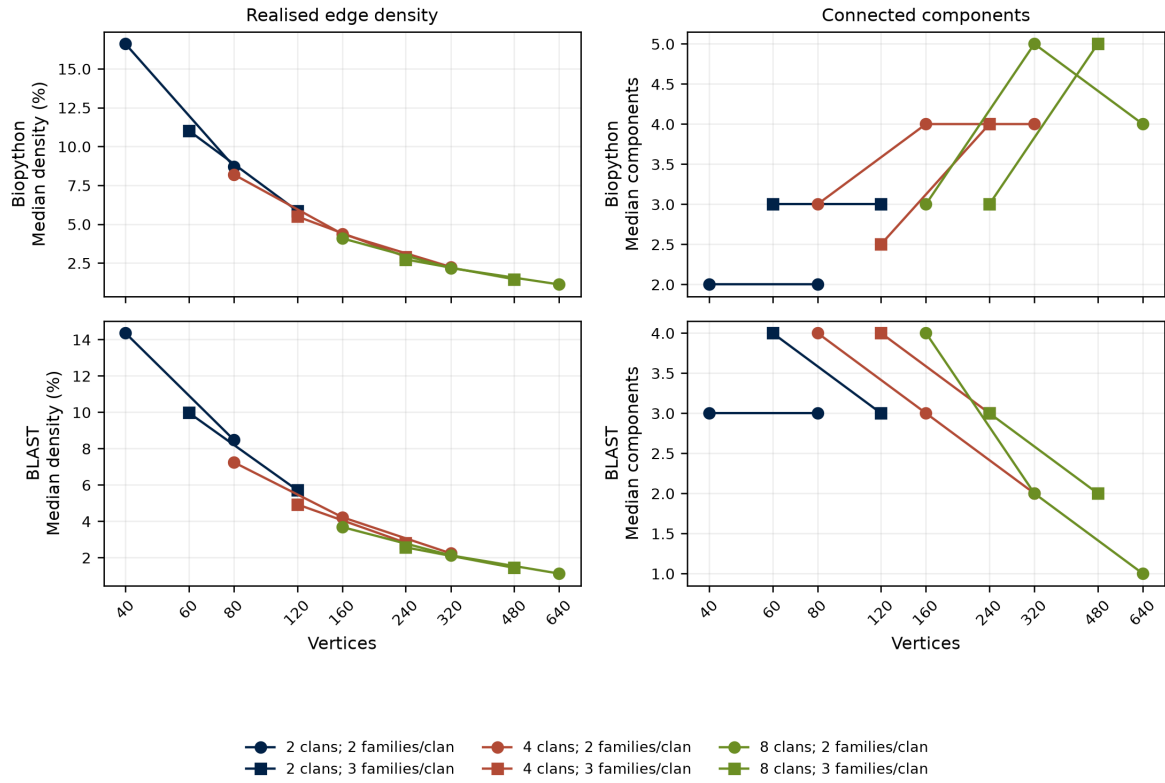


Figure 11: Top-5 graph structure across 14 primary composition conditions. Rows separate score methods. Colour gives clan count; marker gives families per clan. Lines connect domain counts.

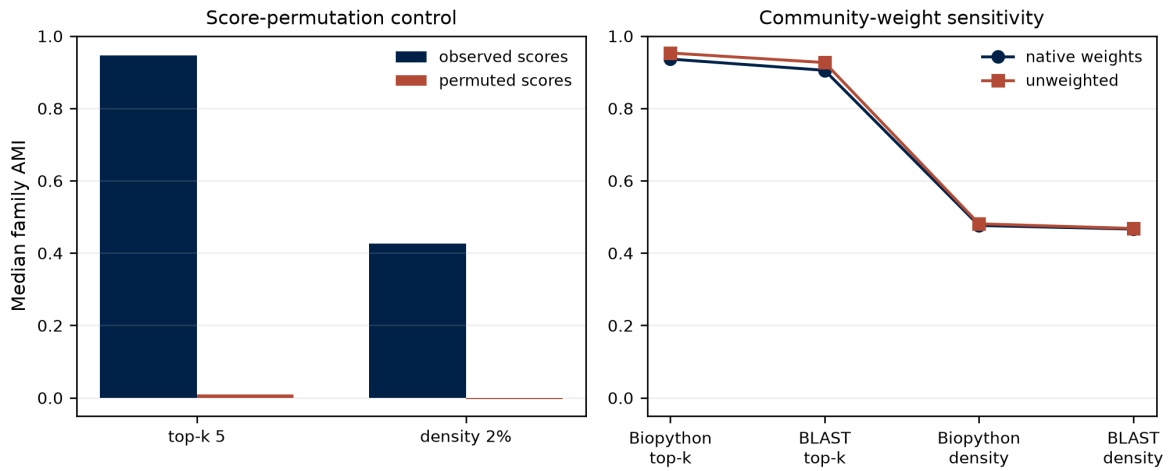


Figure 12: Sensitivity to score permutation and community weight.

5 Discussion

Top- k rules recovered withheld Pfam labels most consistently across tested score methods and label levels. Top-10 maximised median family AMI for both methods and was within 0.006 of highest clan AMI. Top-5 retained fewer edges with median family AMI 0.938 for Biopython and 0.917 for BLAST. Selection between top-5 and top-10 depends on desired graph sparsity.

Edge allocation explains much of this result. Sparse global rules produced many components; top- k produced fewer components at similar density because each vertex selected local neighbours. Global density alone did not determine graph structure.

NMI exceeded AMI under sparse, fragmented settings (Figs. 5 and 6). AMI subtracts expected mutual information conditional on observed group sizes. Comparisons across rules and compositions therefore use AMI; NMI supplies an unadjusted bounded descriptor.

Biopython and BLAST ranks agreed closely among pairs reported by BLAST, while BLAST covered 4.21% of master pairs. Equal percentiles combined score method, coverage, and edge-budget effects. Target density removed the edge-budget difference; at 2% density, median absolute family differences were 0.0034 for NMI and 0.0094 for AMI. Agreement between methods after density matching supports graph-rule conclusions shared by both score systems.

Collection composition changed density, component count, and recovery under a fixed top-5 rule. Balanced sampling estimates behaviour across controlled mixtures rather than natural Pfam abundance. Replicate variation within each condition captures domain and family selection effects for this master panel.

Permutation results show that pair identity contains family information beyond score distribution. Under unit weights, family AMI at reference setting was 0.953, compared with 0.936 under native weights. Controls locate observed signal primarily in edge placement for this setting.

Inference scope remains balanced mixtures from 24 selected Pfam families. Full-length proteins, alternative databases, synthetic evolutionary data, and structural labels require separate evaluation. 4 saturated design conditions reduced primary factorial design to 14 conditions. Louvain resolution and stochasticity received limited study. DEDAL lacks scaled evidence here.

Future work should repeat sampling across broader Pfam panels, model composition effects with hierarchical uncertainty, test graph filtrations across continuous edge budgets, and compare community algorithms. These experiments estimate stability across panels and graph construction choices.

6 Conclusion

Pipeline maps recorded domain collections to pair scores, weighted graphs, communities, and auditable agreement measures. Among tested settings, top- k rules recovered withheld Pfam structure most consistently; top-10 maximised family AMI and top-5 supplied a sparser alternative with median family AMI above 0.91 under both score methods. AMI corrected chance agreement caused by variable community counts. Biopython and BLAST produced similar recovery after edge budgets were matched, while equal percentiles produced different graph densities. Sampling frame, score coverage, edge rule, and realised density should accompany each reported SSN result.

A Panel selection algorithm

Parameters used were $c^* = 8$, $f^* = 3$, $r^* = 40$, and $\sigma = 20260828$.

Algorithm 1 Seeded selection of clan and family panel

Require: catalogue of Pfam families with clan assignment; targets c^* clans and f^* families per clan; minimum reported seed count r^* ; seed σ

Ensure: panel of c^*f^* families, and trace of every family inspected

```
1:  $C \leftarrow$  clans holding at least  $f^*$  catalogue families, excluding unassigned
2: sort  $C$ , then shuffle under  $\sigma$ 
3:  $P \leftarrow \emptyset$ 
4: for all  $C \in C$  in shuffled order do
5:   sort families of  $C$ , then shuffle under same stream
6:    $A \leftarrow \emptyset$ 
7:   for all  $F$  in shuffled families of  $C$  do
8:     query InterPro for entry type and reported seed count of  $F$ 
9:     append  $F$  to trace
10:    if type of  $F$  is domain or family, and seed count  $\geq r^*$  then
11:       $A \leftarrow A \cup \{F\}$ 
12:    end if
13:    if  $|A| = f^*$  then
14:      break
15:    end if
16:  end for
17:  if  $|A| = f^*$  then
18:     $P \leftarrow P \cup A$  ▷ otherwise clan  $C$  is skipped
19:    if  $P$  spans  $c^*$  clans then
20:      break
21:    end if
22:  end if
23: end for
24: return  $P$  and trace
```

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